



Gene Expression-Based Cancer Classification

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Objectives

- Classify cancer types using gene expression data
- Reduce dimensionality using feature selection
- Compare multiple ML models
- Evaluate performance using standard metrics

Introduction

Recent advances in bioinformatics have enabled the analysis of large-scale gene expression data, providing valuable insights into the biological mechanisms of cancer. Gene expression profiles capture complex patterns that differ across cancer types, but the high dimensionality of this data makes manual analysis impractical. Machine learning techniques offer powerful tools for automatically identifying relevant patterns within such data. In this project, machine learning methods are applied to gene expression data in order to accurately classify samples into different cancer types.



Motivation & Problem Statement

Motivation

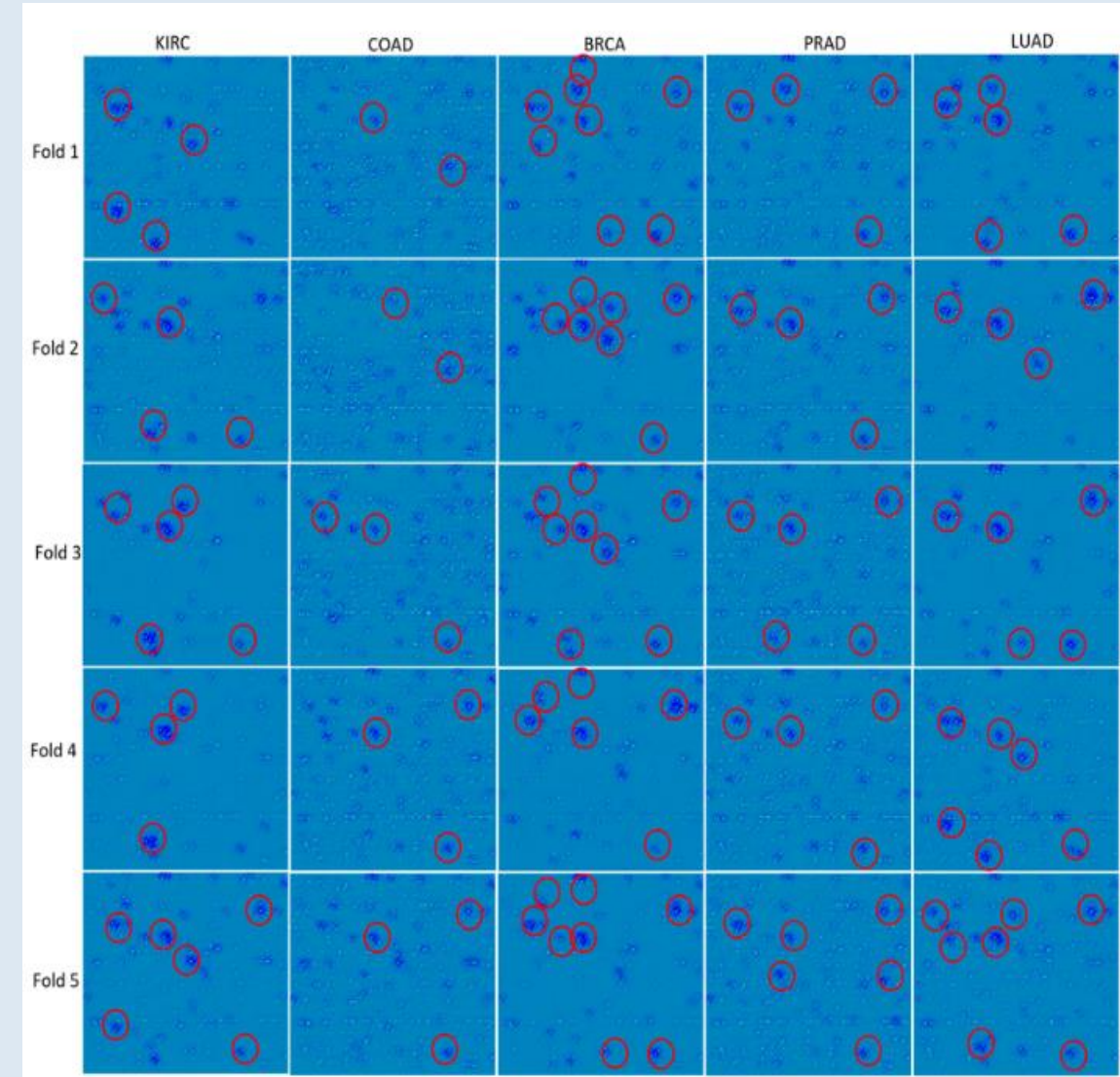
- Cancer diagnosis relies on the analysis of complex biological data.
- Gene expression datasets contain thousands of genes for each patient sample.
- Manual analysis of such high-dimensional data is time-consuming and prone to error.
- Machine learning techniques can automatically learn patterns from gene expression data.

Problem Statement

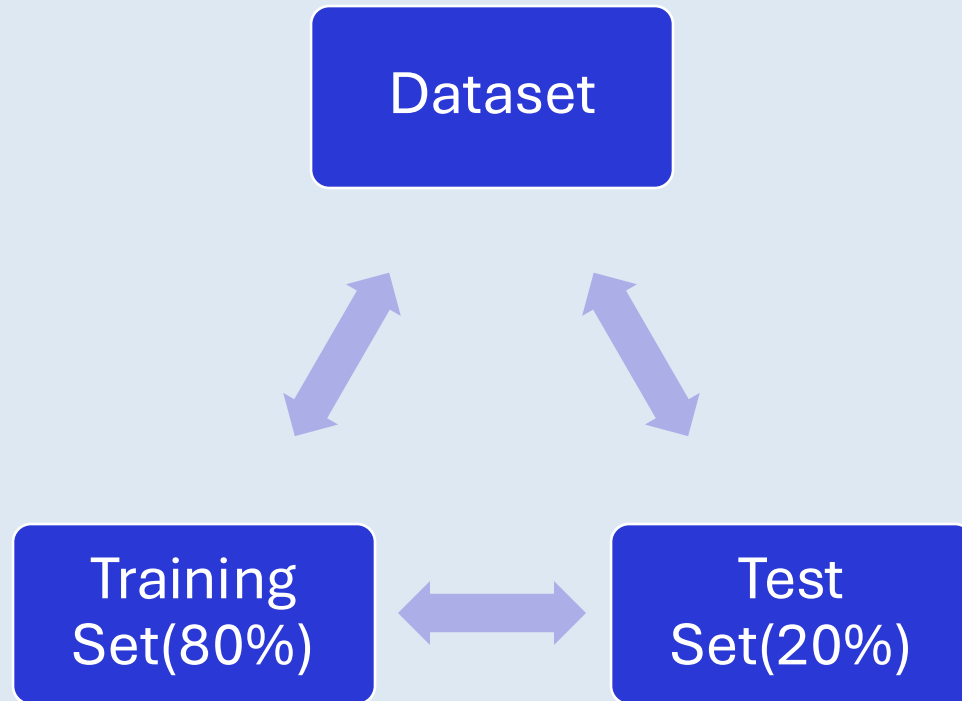
- Given gene expression measurements, accurately classify samples into different cancer types.
- Handle the challenge of very high dimensionality (tens of thousands of gene features).
- Identify informative genes while maintaining high classification accuracy.

Dataset Overview

- The dataset used in this project consists of gene expression measurements collected from cancer patient samples. Each sample is represented by a large number of numerical features corresponding to gene expression levels. The dataset contains a total of 801 samples, with approximately 20,531 gene features per sample. Each sample is labeled as one of five cancer types:
- BRCA
- KIRC
- LUAD
- PRAD
- COAD.
- The class distribution is imbalanced, with BRCA having the largest number of samples and COAD the smallest. This high-dimensional and multi-class structure makes the dataset well suited for evaluating machine learning classification methods.



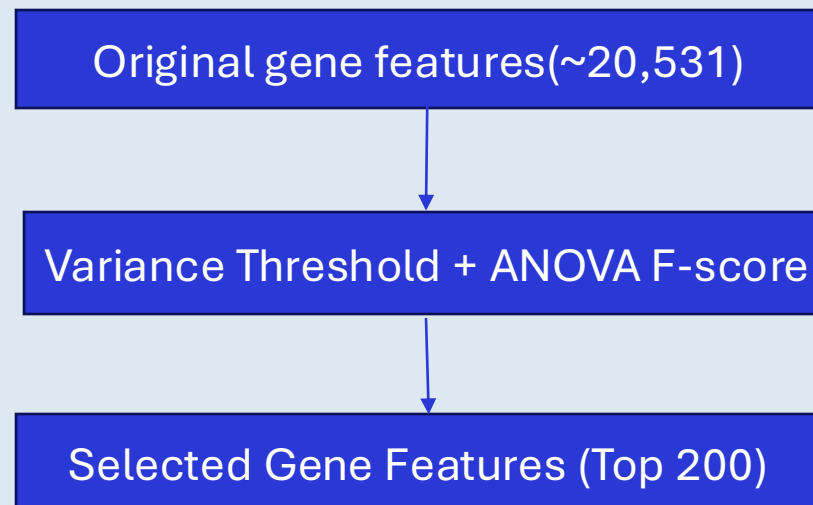
Data Preprocessing



Before training the machine learning models, several preprocessing steps were applied to ensure data quality and reliable evaluation. Only numerical gene expression features were retained, as required by the learning algorithms. The dataset was divided into training and testing sets using an 80/20 split, with stratification applied to preserve the original class distribution across cancer types. This approach ensures that all classes are fairly represented during training and evaluation. No missing or duplicate samples were present in the dataset, allowing the models to be trained directly on the processed data.

Feature Selection Strategy

Gene expression data is highly dimensional, with tens of thousands of gene features for each sample. Using all features directly can lead to overfitting and increased computational cost. To address this challenge, feature selection was applied to reduce dimensionality while preserving discriminative information. A variance threshold was first used to remove genes with little variation across samples. Subsequently, the ANOVA F-score method was applied to rank features based on their relevance to cancer class labels, and the top 200 genes were selected. This approach improves model performance, efficiency, and interpretability.



Machine Learning Models Used

Logistic Regression

A linear classification model used as a strong and interpretable baseline for multi-class cancer classification.

Support Vector Machine (Linear Kernel)

A powerful classifier well suited for high-dimensional data, designed to maximize class separation in feature space.

K-Nearest Neighbors (KNN)

A distance-based classifier that assigns class labels based on the majority class among the nearest neighbors.

Naive Bayes

A probabilistic baseline model based on Bayes' theorem, assuming conditional independence between gene features.

Results & Evaluation

The performance of the machine learning models was evaluated using

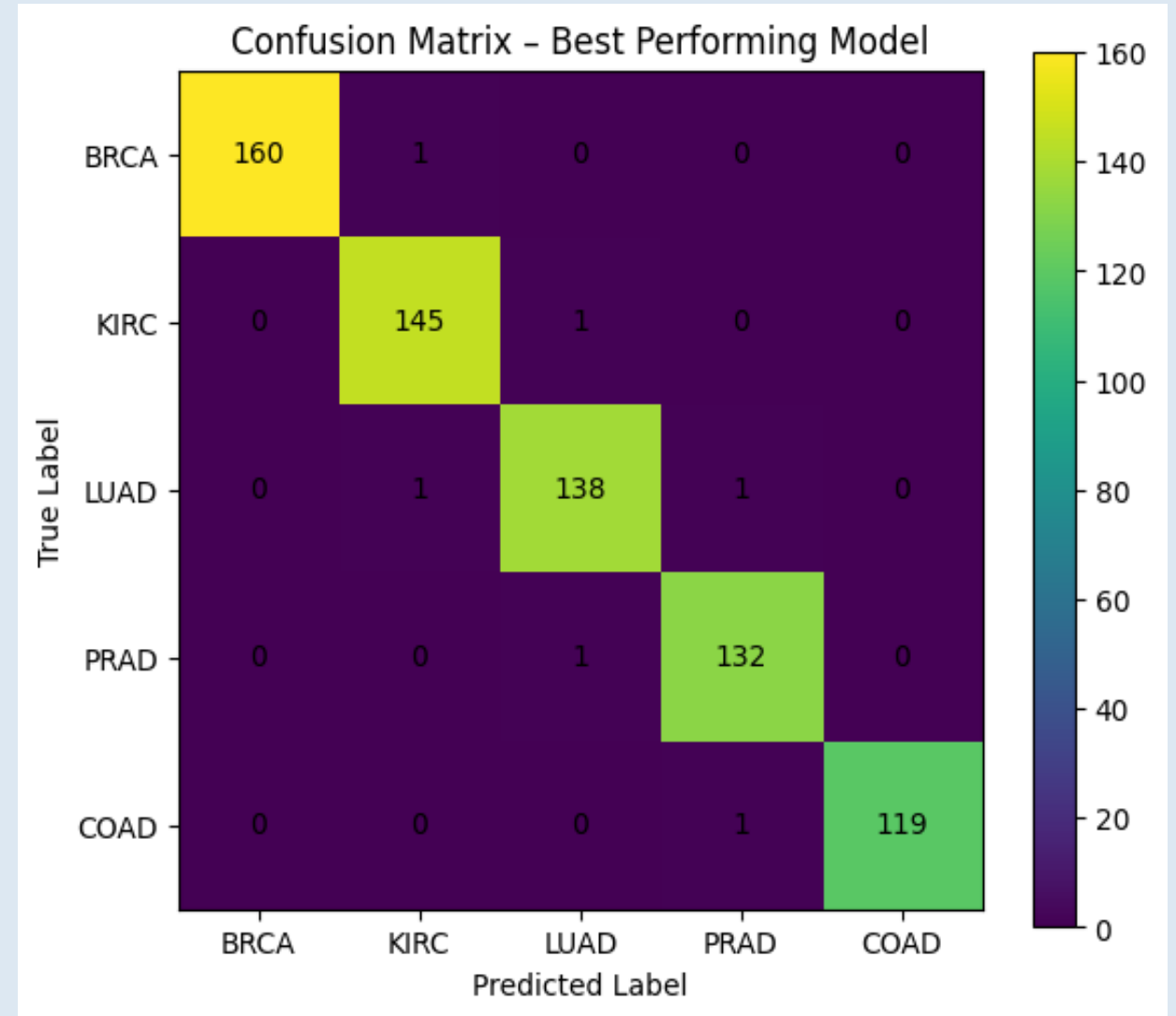
- accuracy and
- macro-averaged F1-score
- to ensure fair comparison across all cancer classes.

Linear models achieved the highest performance, with Logistic Regression and Linear SVM both reaching approximately 99% accuracy and macro F1-score. K-Nearest Neighbors also performed well, while Naive Bayes showed significantly lower performance. These results indicate that linear classifiers are highly effective for high-dimensional gene expression data when combined with appropriate feature selection.

Model	Accuracy	Macro F1
Logistic Regression	~0.994	~0.994
SVM(Linear)	~0.994	~0.994
KNN	~0.976	~0.977
Naïve Bayes	~0.807	~0.756

Confusion matrix

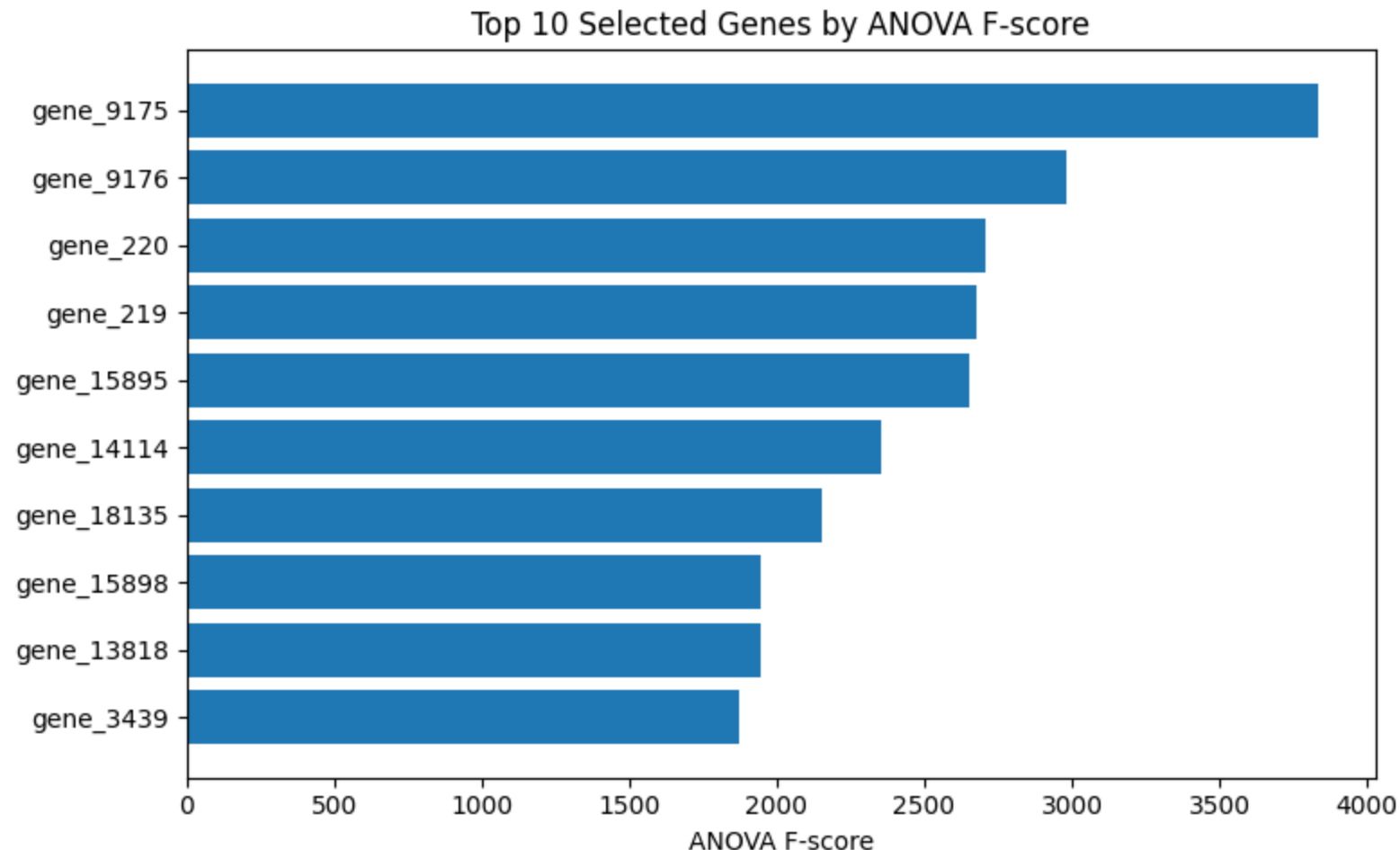
The confusion matrix of the best performing model illustrates the classification performance across all cancer types. Most samples are correctly classified along the main diagonal, indicating a very high level of accuracy. Only a small number of misclassifications are observed, primarily between biologically similar cancer types. This result confirms that the model is able to effectively distinguish between different cancer classes based on gene expression data.



Feature Importance (Biological Insight)

Feature selection enables the identification of genes that contribute most to cancer classification. Using the ANOVA F-score, the most discriminative genes were ranked based on their statistical association with cancer class labels.

The displayed features represent the top-ranked genes, highlighting those that play a key role in distinguishing between different cancer types. This analysis provides biological insight into the gene expression patterns learned by the model.



Limitations & Future Work

Limitations

- Limited dataset size compared to the high number of gene features.
- Lack of biological validation for the selected genes.
- Evaluation performed on a single dataset only.
- Linear models may not capture complex non-linear gene interactions.
- Clinical and patient metadata were not included in the analysis.

Future Work

- Evaluate the model on larger and more diverse gene expression datasets.
- Perform biological validation of the most informative genes.
- Explore advanced feature selection and regularization methods.
- Investigate deep learning models to capture non-linear gene interactions.
- Integrate gene expression data with clinical information for improved prediction.

Conclusion

This project demonstrated that machine learning can effectively classify cancer types using gene expression data. Feature selection was essential due to the very high dimensionality of the dataset, and the combination of variance thresholding with ANOVA F-score successfully reduced the feature space while preserving the most informative genes.

Among the evaluated models, **Logistic Regression and Linear Support Vector Machine** achieved the best performance. These models were chosen because they are well suited for high-dimensional data and provide stable and interpretable decision boundaries. After feature selection, the data exhibited strong linear separability, which explains the superior performance of these linear classifiers. Overall, the results confirm that appropriate feature selection combined with simple yet robust models can achieve high accuracy in gene expression-based cancer classification.

Thank you for your attention!